

Sesi 7

21 April 2026

A Macroeconomic Theory of the Open Economy

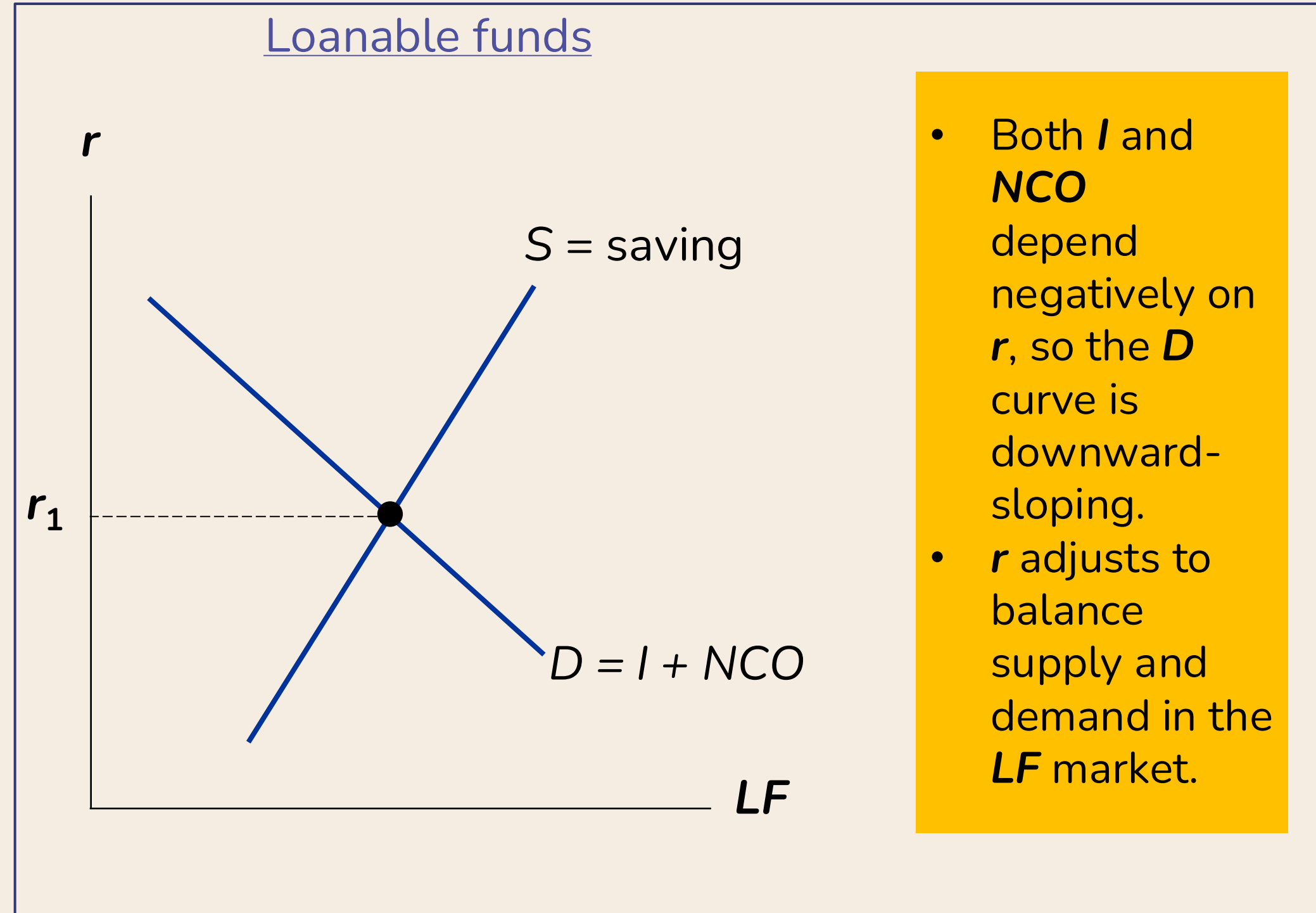
Pengantar Ekonomi 2

Sumber: Mankiw (Ch 33)

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The Market for Loanable Funds (1)

- In an open economy, $S = I + NCO$
 - Saving = Domestic investment + Net capital outflow
- Market for loanable funds:
 - Supply of loanable funds: from national saving (S)
 - Demand for loanable funds: from domestic investment (I) and net capital outflow (NCO)
- Net outflow of capital, when $NCO > 0$
 - Net purchase of capital overseas adds to the demand for domestically generated loanable funds
- Net inflow of capital, when $NCO < 0$
 - Capital resources coming from abroad reduce the demand for domestically generated loanable funds



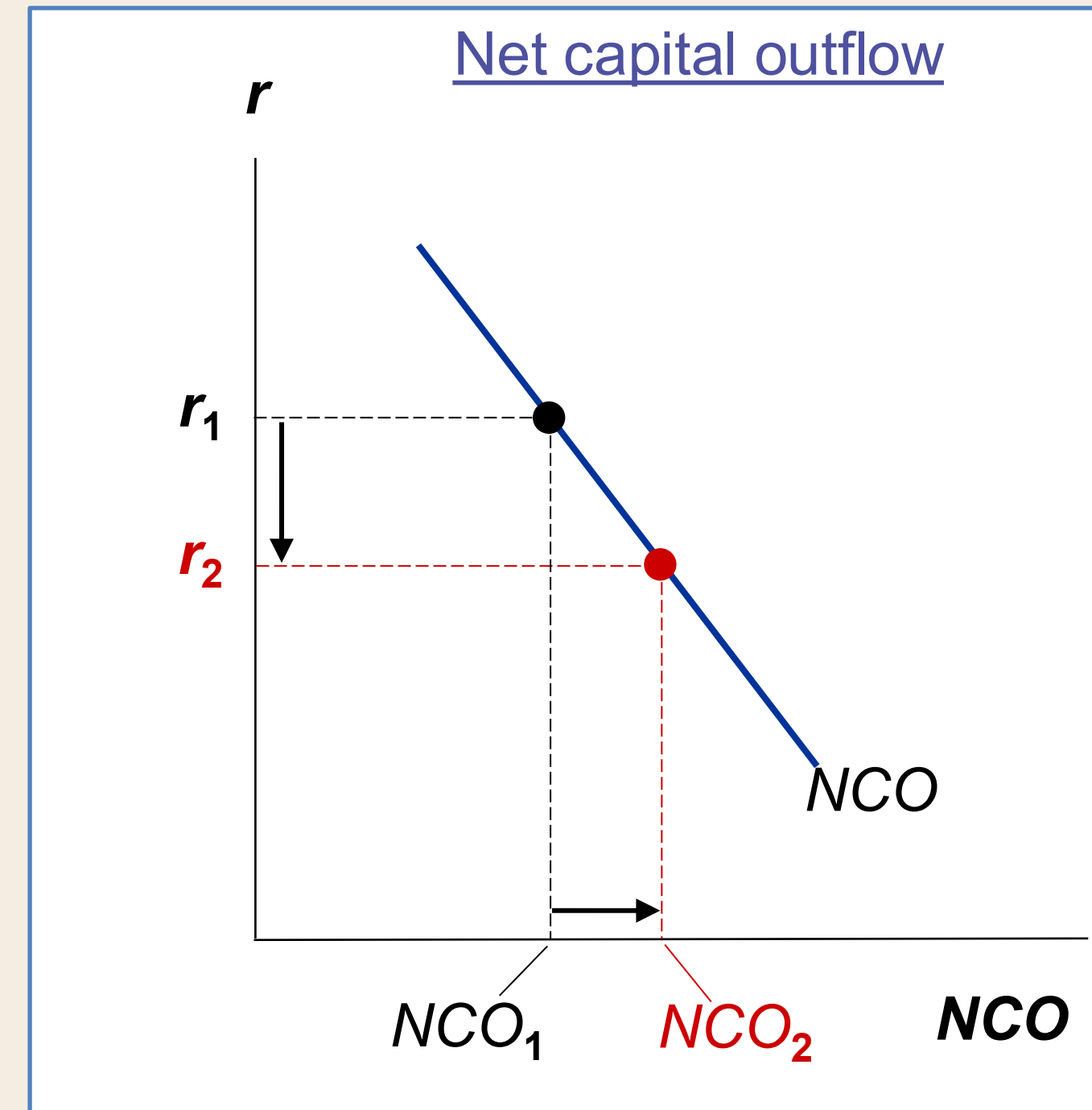
The Market for Loanable Funds (2)

How NCO Depends on the Real Interest Rate

The real interest rate, r , is the real return on domestic assets.

A fall in r makes domestic assets less attractive relative to foreign assets.

- People in the U.S. purchase more foreign assets.
- People abroad purchase fewer U.S. assets.
- **NCO** rises.



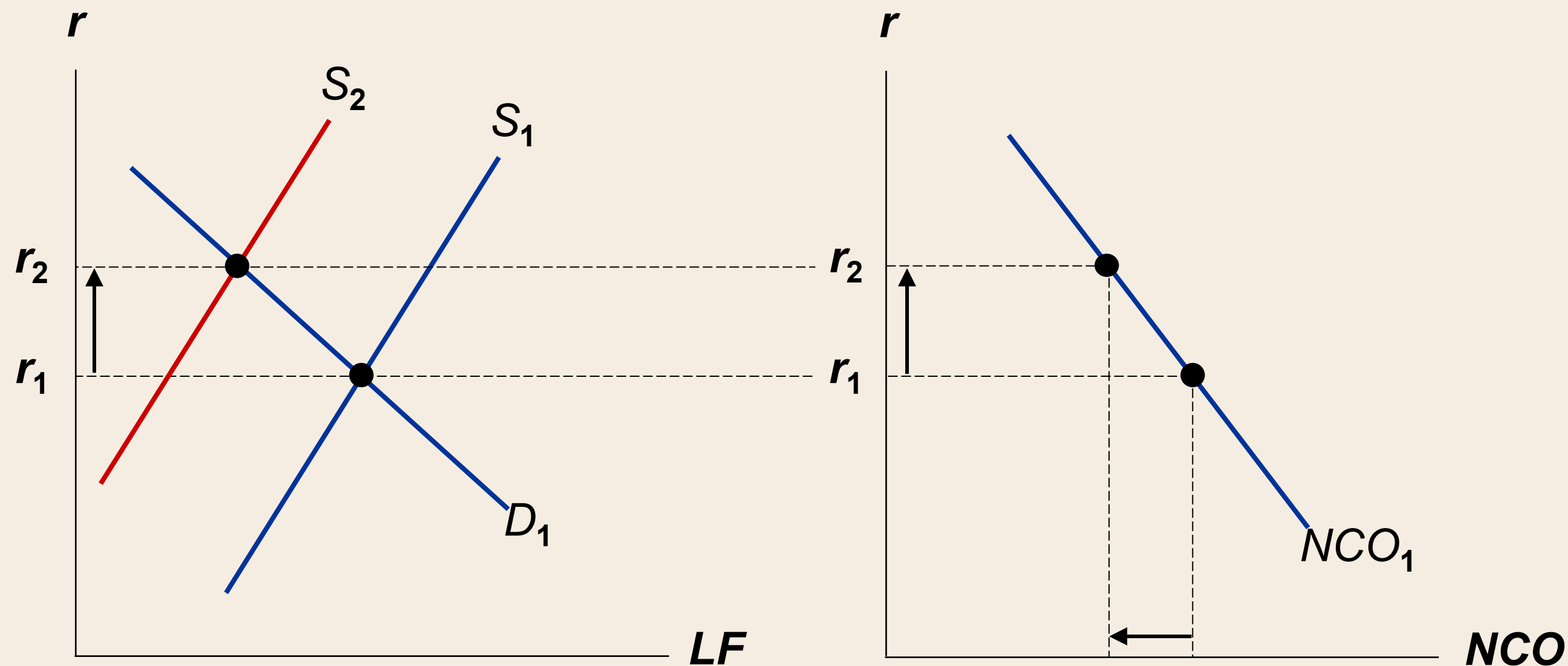
EXAMPLE 1: Budget deficits and capital flows

Suppose the government runs a budget deficit (previously, the budget was balanced).

- Use the appropriate diagrams to determine the effects on the real interest rate and net capital outflow.

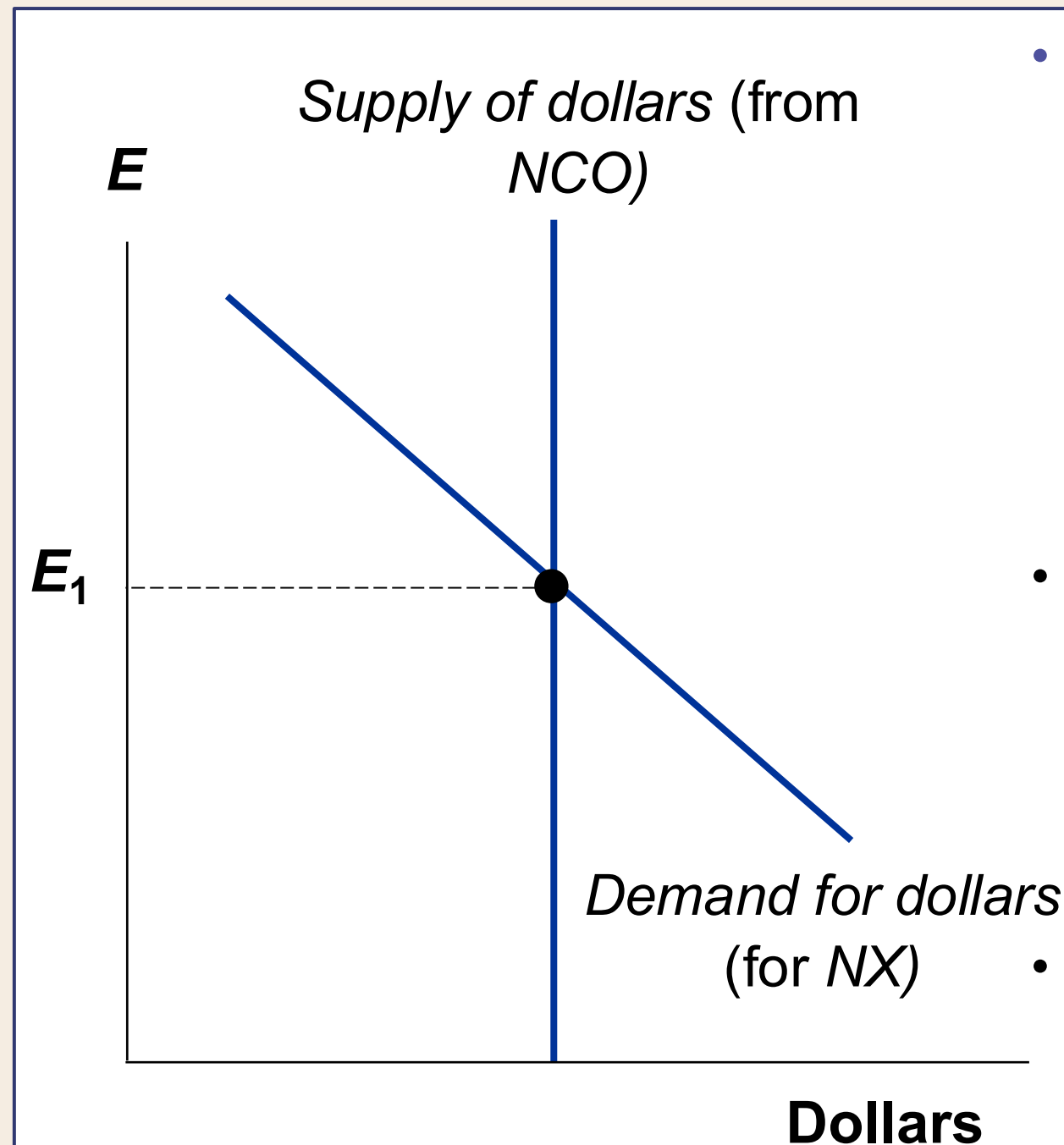
Solution:

When working with this model, keep in mind:
the LF market determines r (in left graph),
then this value of r determines NCO (in right graph).



Foreign-Currency Exchange Market

- The market for foreign-currency exchange
 - Trade U.S. dollars in exchange for foreign currencies
 - Identity: $NCO = NX$
 - NX is the demand for dollars: foreigners need dollars to buy U.S. net exports.
 - NCO is the supply of dollars: U.S. residents sell dollars to obtain the foreign currency they need to buy foreign assets.
- The U.S. real exchange rate (E)
 - Measures the quantity of foreign goods & services that trade for one unit of U.S. goods & services.
 - E is the real value of a dollar in the market for foreign-currency exchange.
 - E balances the supply and demand in the market for foreign-currency exchange



- An increase in E makes U.S. goods more expensive to foreigners and reduces the quantity of dollars demanded to buy those goods.
- An increase in E has no effect on S or I , so it does not affect NCO or the supply of dollars.
- E adjusts to balance supply and demand for dollars in the market for foreign-currency exchange.

Active Learning 1: Budget deficits

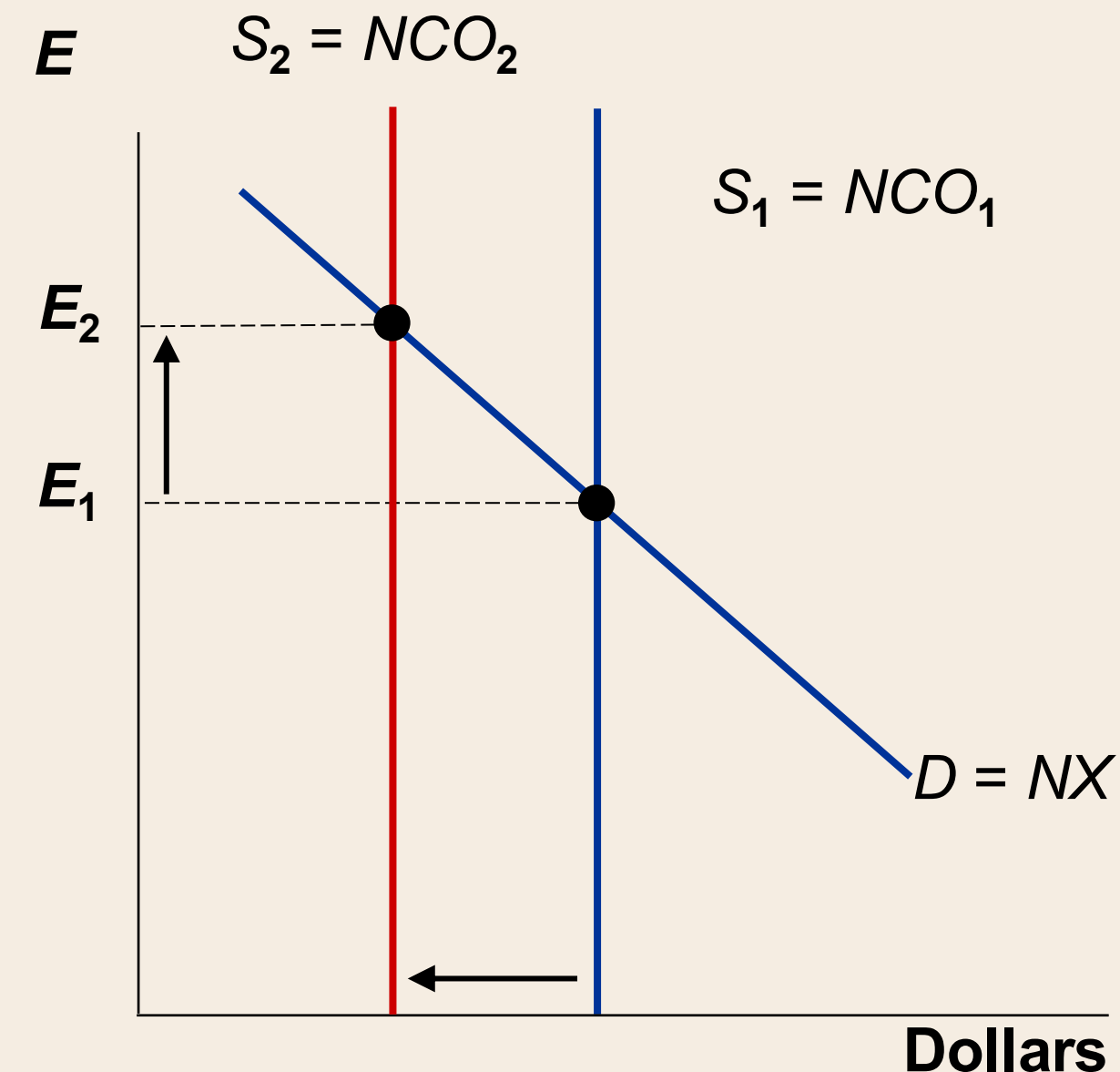
Initially, the government budget is balanced, and trade is balanced ($NX = 0$). Suppose the government runs a budget deficit. As we saw earlier, r rises and NCO falls.

- How does the budget deficit affect the U.S. real exchange rate? (graph the foreign-exchange market)
- How does the budget deficit affect the U.S. balance of trade?

Answers:

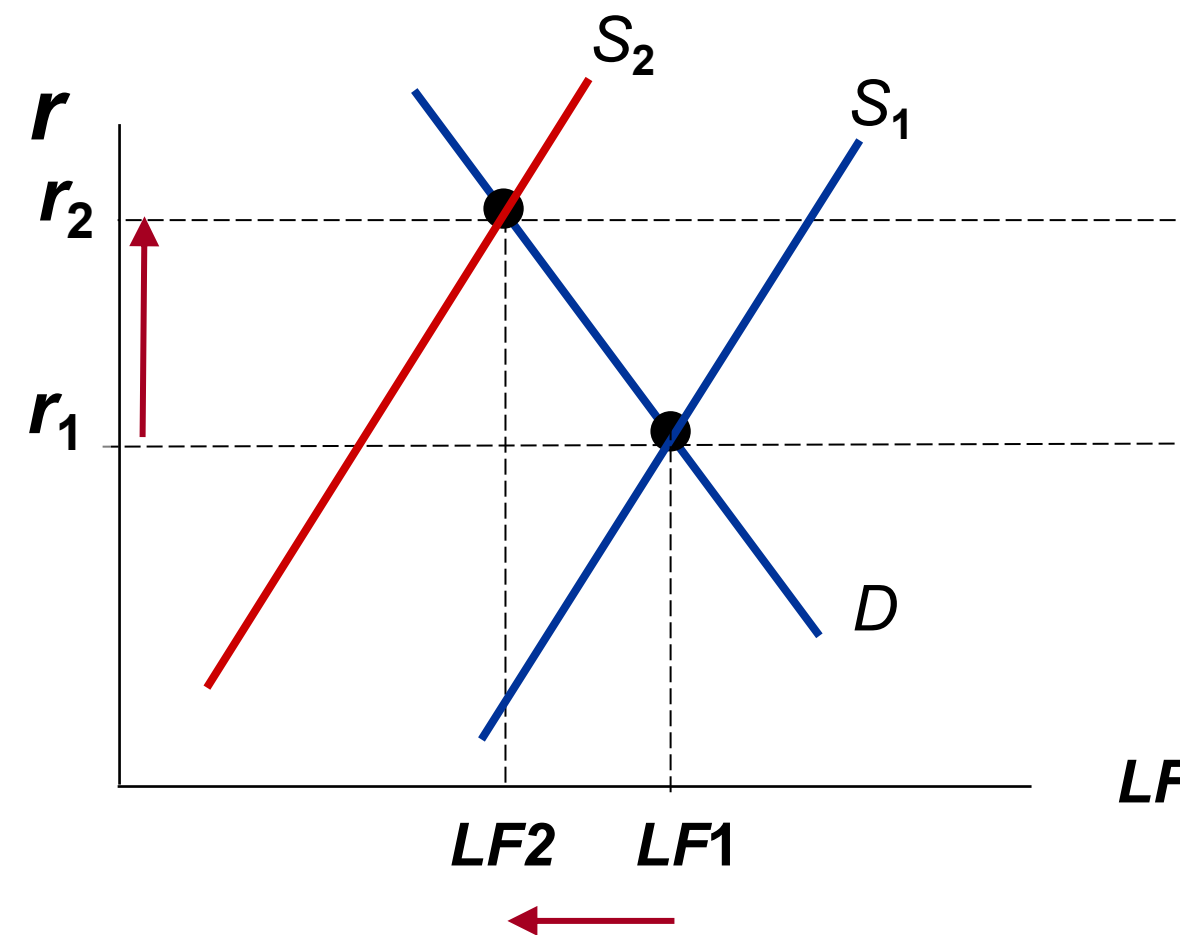
The effects of a budget deficit:

- National saving falls (S), the real interest rate rises, reduces NCO and the supply of dollars.
- The real exchange rate appreciates, reducing net exports.
- Since $NX = 0$ initially, the budget deficit causes a trade deficit ($NX < 0$).

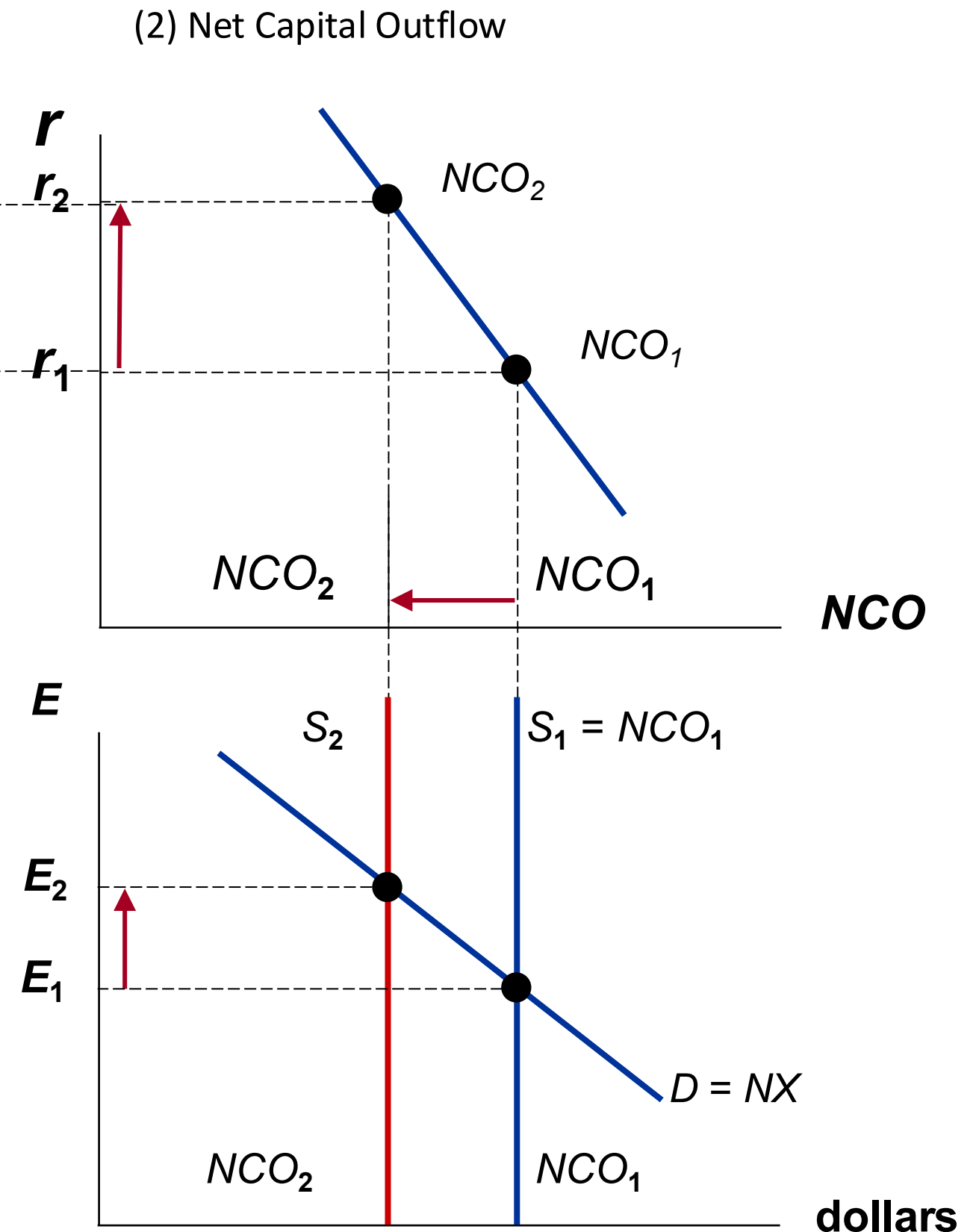


The Connection Between r and e

- Anything that increases r will reduce **NCO** and the supply of dollars in the foreign exchange market.
- Result: The real exchange rate appreciates.
- Keep in mind: The **LF** market determines r .
- This r determines **NCO**
This **NCO** determines supply of dollars in foreign exchange market.



(1) Market for Loanable Fund



(3) Market for Foreign Exchange

Active Learning 2: Investment incentives

Suppose the government provides new tax incentives to encourage investment.

- Use the appropriate diagrams to determine how this policy would affect:
 - A. the real interest rate, r
 - B. net capital outflow, NCO
 - C. the real exchange rate, E
 - D. net exports, NX

Trade Policy

- Trade policy:
 - Government policy that directly influences the quantity of goods and services a country imports or exports
 - Tariff: a tax on imported goods
 - Import quota: limit on the quantity of imports
- Some arguments for restricting trade:
 - Save jobs in domestic industry
 - Reduce the trade deficit

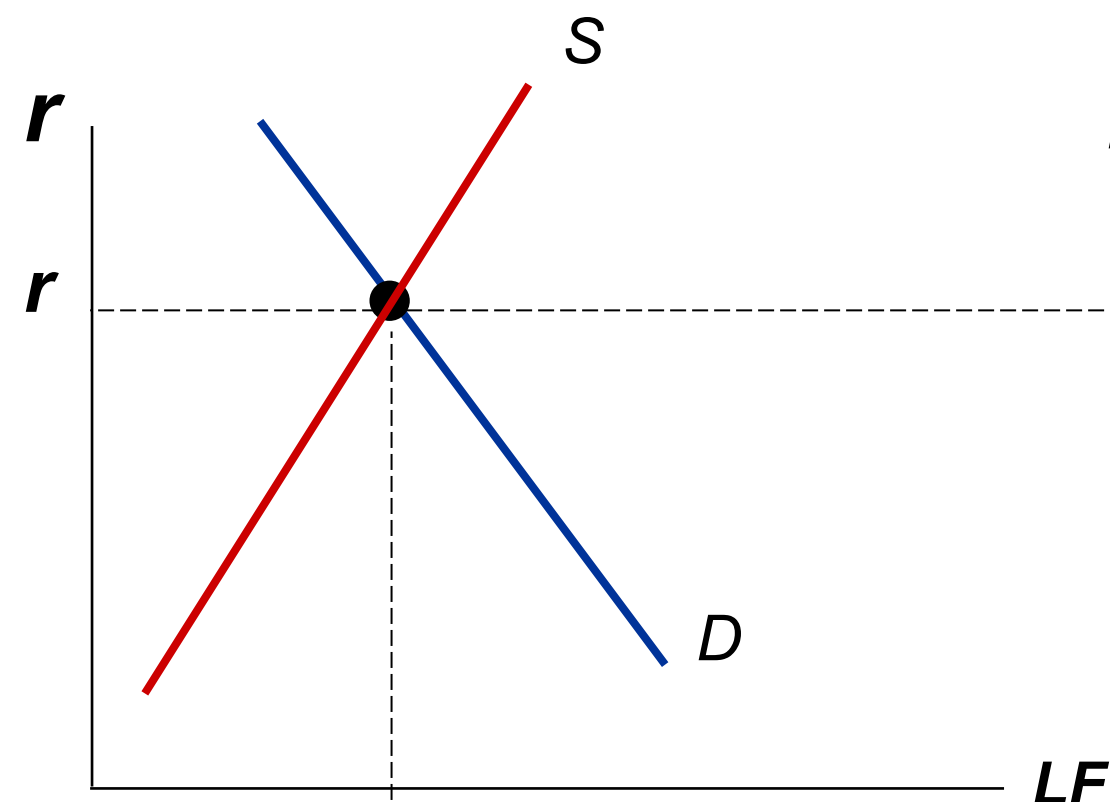
EXAMPLE 2: Protecting domestic auto makers

Suppose the government uses import quotas on cars imported from Japan, in order to protect jobs in the domestic auto industry.

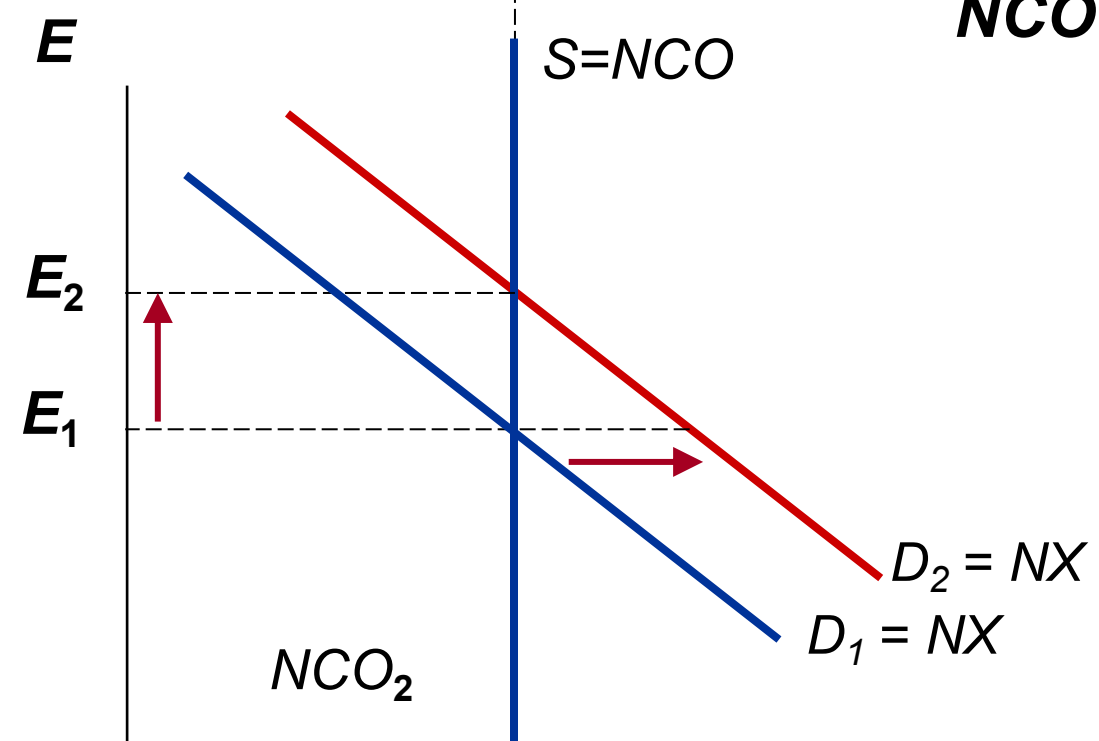
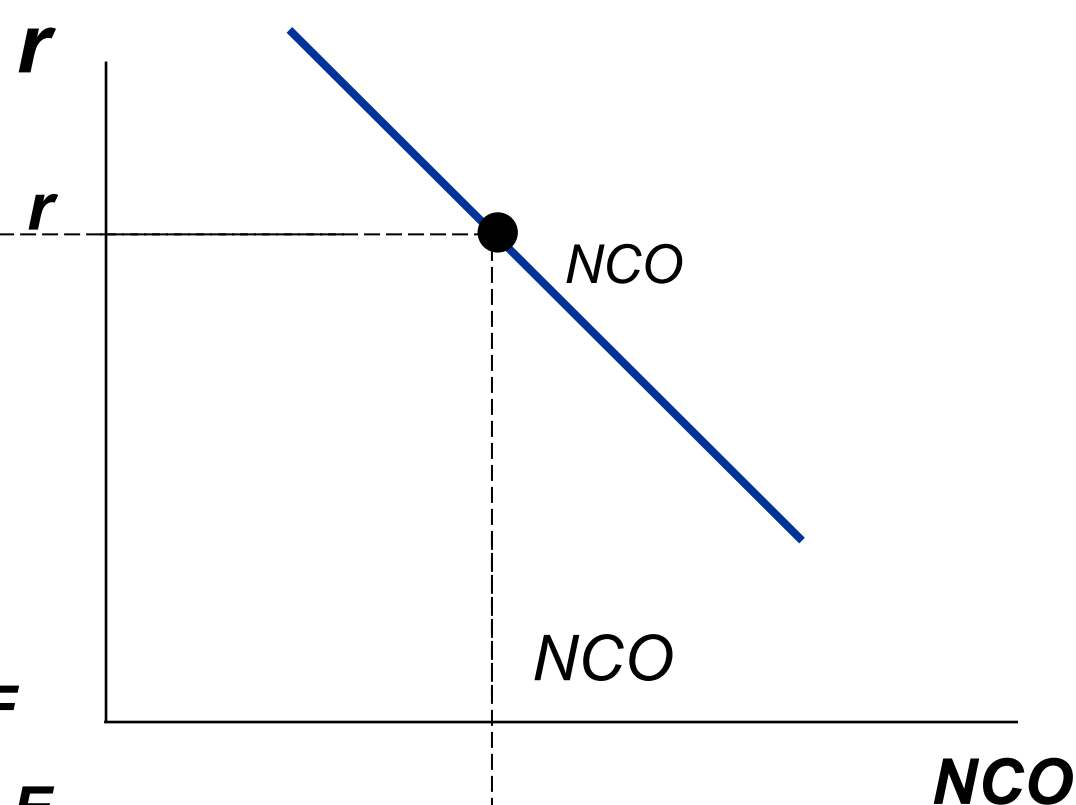
- Use the appropriate diagrams to determine how this policy would affect:
 - A. the real interest rate, r
 - B. net capital outflow, NCO
 - C. the real exchange rate, E
 - D. net exports, NX
 - E. Does the policy save U.S. jobs?

Trade Policy

(1) Market for Loanable Fund



(2) Net Capital Outflow



(3) Market for Foreign Exchange **Dollar**

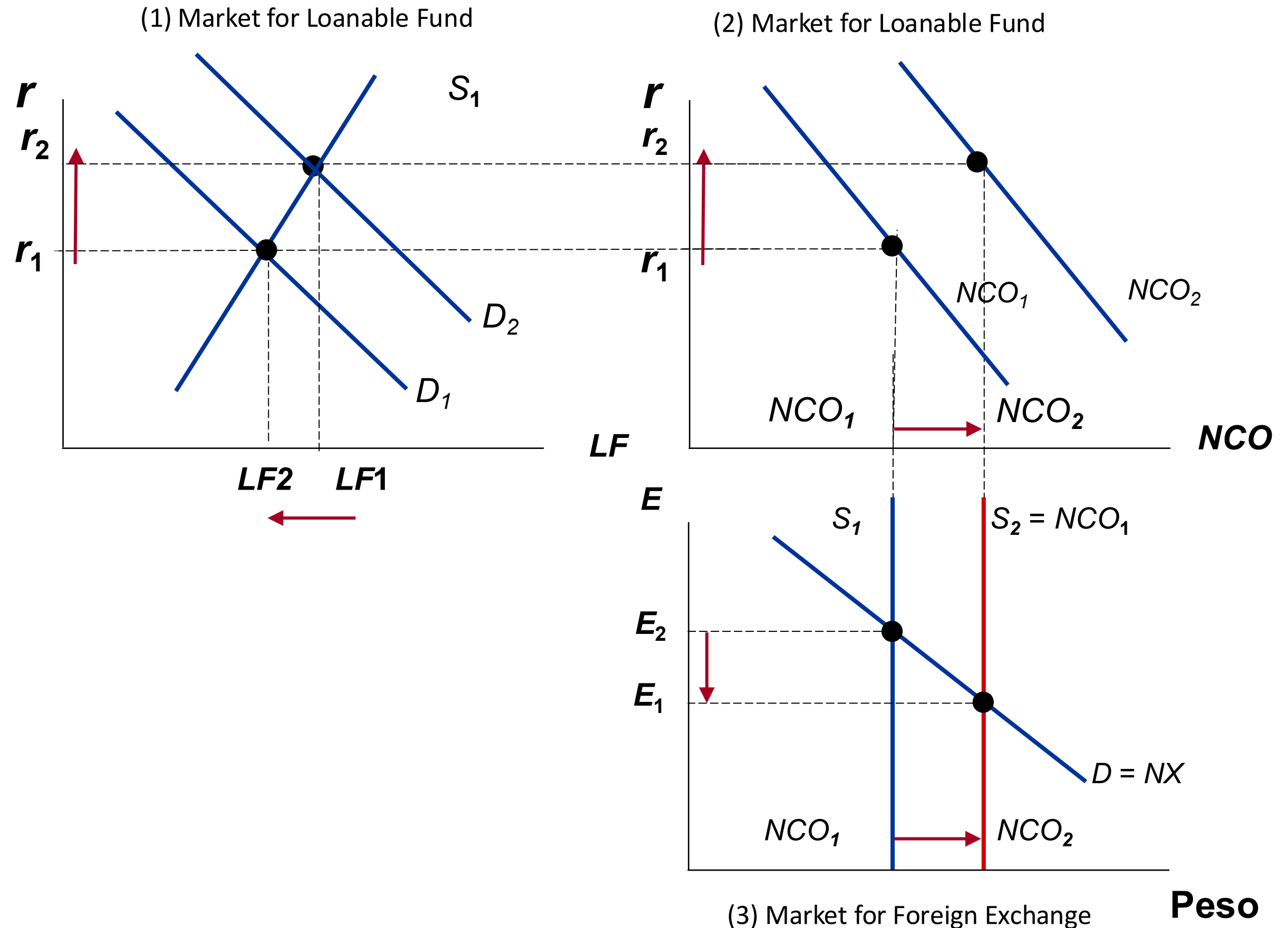
Case: Import Quota

- An import quota does not affect saving or investment, so, it does not affect **NCO**, S curve does not shift.
- At each E , imports of cars fall, so net exports rise, D shifts to the right.
- At E_1 , there is excess demand in the foreign exchange market. E rises to restore equilibrium.
- What happens to **NX**? Nothing!
 - If E could remain at E_1 , **NX** would demanded would rise.
 - But the import quota does not affect **NCO**, so the quantity of dollars supplied is fixed.
 - Since **NX** must equal **NCO**, E must rise enough to keep **NX** at its original level.
 - *Hence, the policy of restricting imports does not reduce the trade deficit.*
- Does the policy save jobs?
 - *The import quota saves jobs in the auto industry but destroys jobs in U.S. export industries!!*

Political Instability and Capital Flight

Case: Capital flight from Mexico

- As foreign investors sell their assets and pull out their capital, NCO increases at each value of r .
- Demand for $LF = I + NCO$. The increase in NCO increases demand for LF .
- The equilibrium values of r and NCO both increase.
- The increase in NCO causes an increase in the supply of pesos in the foreign exchange market.
- The real exchange rate value of the peso falls.



CHAPTER IN A NUTSHELL

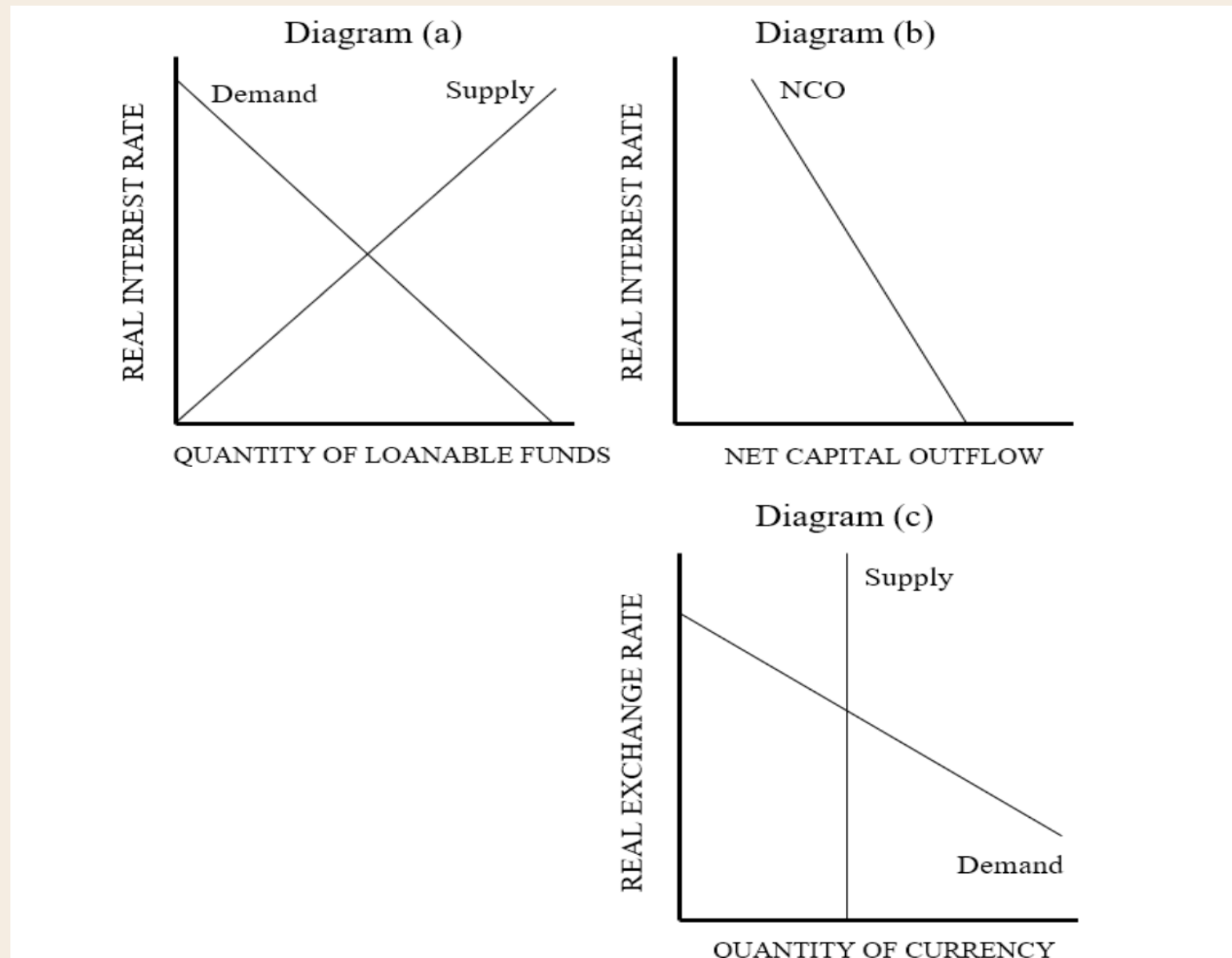
- A policy that reduces national saving, such as a government budget deficit, reduces the supply of loanable funds and drives up the interest rate. It causes net capital outflow to decline, reducing supply of dollars in the market for foreign-currency exchange. The dollar appreciates, and net exports fall.
- Political instability can lead to capital flight, which tends to increase interest rates and cause the currency to depreciate.
- Although restrictive trade policies, such as tariffs or quotas on imports, are sometimes advocated as a way to alter the trade balance, they do not necessarily have that effect. A trade restriction increases net exports for any given exchange rate and, therefore, increases the demand for dollars in the market for foreign-currency exchange. As a result, the dollar appreciates, making domestic goods more expensive relative to foreign goods. This appreciation offsets the initial impact of the trade restriction on net exports.

Latihan Soal

1. Kondisi yang menyebabkan terjadinya depresiasi nilai tukar riil adalah
 - a. kenaikan defisit anggaran
 - b. pelarian modal ke luar negeri
 - c. kebijakan pembatasan impor
 - d. kebijakan kontraksi moneter

2. Dengan asumsi hal-hal lain tak berubah (*ceteris paribus*), jika keseimbangan tingkat bunga riil di Amerika Serikat (AS) meningkat, maka arus keluar modal netto (*NCO*) di AS akan
 - a. meningkat, dan arus keluar modal netto (*NCO*) negara lain juga akan meningkat
 - b. meningkat, sementara arus keluar modal netto (*NCO*) negara lain akan menurun
 - c. menurun, dan arus keluar modal netto (*NCO*) negara lain akan menurun
 - d. menurun, sementara arus keluar modal netto (*NCO*) negara lain akan meningkat

3. Gunakan diagram 3 (tiga) panel model makroekonomi perekonomian terbuka di bawah ini untuk menjawab pertanyaan berikut.



Misalkan pada Tahun 2025 Pemerintah Qatar mengubah kebijakan anggarannya dari defisit anggaran menjadi surplus anggaran. Dampak dari perubahan ini dapat digambarkan oleh

- pergeseran kurva permintaan pada panel (a) ke kiri dan kurva penawaran pada panel (c) ke kiri
- pergeseran kurva permintaan pada panel (a) ke kanan dan kurva permintaan pada panel (c) ke kiri
- pergeseran kurva penawaran pada panel (a) ke kanan dan kurva penawaran pada panel (c) ke kanan
- pergeseran kurva penawaran pada panel (a) ke kanan dan kurva permintaan pada panel (c) ke kanan

4. Apabila Amerika Serikat menaikkan bea masuk impor mobil, maka pada pasar nilai tukar terjadi
- a. kelebihan penawaran di pasar pertukaran mata uang asing, sehingga nilai tukar riil akan terapresiasi
 - b. kelebihan penawaran di pasar pertukaran mata uang asing, sehingga nilai tukar riil akan terdepresiasi
 - c. kelebihan permintaan di pasar pertukaran mata uang asing, sehingga nilai tukar riil akan terapresiasi
 - d. kelebihan permintaan di pasar pertukaran mata uang asing, sehingga nilai tukar riil akan terdepresiasi
5. Nilai tukar riil (dalam jangka panjang) tidak dipengaruhi oleh
- a. ekspor netto
 - b. jumlah uang beredar
 - c. nilai tabungan dan/atau investasi nasional
 - d. arus modal keluar netto

Esai:

1. Venezuela adalah salah satu negara yang mengalami masalah ekonomi paling parah dalam beberapa tahun terakhir. Krisis ekonomi berkepanjangan yang diperparah oleh ketidakstabilan politik dan penurunan harga energi, telah melemahkan mata uang dan sistem perbankan negara tersebut. Banyak warga Venezuela memilih untuk memindahkan uang mereka ke negara-negara Amerika Latin lainnya atau negara-negara lain yang lebih aman, yang mengakibatkan pelarian modal (*capital flight*) yang signifikan. (Sumber: BBC, 2024).

Sebagai seorang Ekonom, buatlah analisis bagaimana krisis tersebut memengaruhi nilai tukar, *net capital outflow* (NCO), dan *loanable funds* di Venezuela. Gunakan diagram yang relevan untuk mendukung analisis Anda.

2. Misalkan seorang kandidat presiden berjanji untuk meningkatkan surplus anggaran pemerintah dan mengklaim bahwa hal tersebut akan menghentikan warga Amerika Serikat berinvestasi di perusahaan asing serta meningkatkan nilai dolar. Evaluasilah janji kandidat tersebut.

3. Micronesia merupakan sebuah negara perekonomian terbuka kecil. Berikut adalah data-data yang dimiliki oleh negara tersebut (dalam miliar).

PDB: **\$7.400**

Konsumsi: **\$4.550**

Pengeluaran pemerintah: **\$1.200**

Ekspor neto: – **\$750**

- i. Berdasarkan data tersebut, hitunglah besarnya nilai investasi, tabungan nasional dan *net capital outflow* (NCO) dari Micronesia.
- ii. Cali adalah seorang turis dari Micronesia yang sedang berlibur di negara Macronesia. Saat liburan, Cali membeli sebuah topi *vintage* di Macronesia seharga 45 Ma\$ (Macro Dollar). Topi yang sama memiliki harga sebesar 39.99 Mi\$ (Micro Dollar) di negaranya. Nilai tukar yang berlaku pada saat itu adalah $\text{Mi\$1} = \text{Ma\$1,12}$. Menurut Anda, harga topi tersebut lebih murah di negara mana? Jelaskan.

4. Pada awal bulan Mei 2023, otoritas moneter Amerika Serikat (*the Fed*) telah menaikkan suku bunga (*Fed rate*) untuk kesepuluh kalinya dalam setahun terakhir. Suku bunga *Fed* yang mencapai 5,25% adalah yang tertinggi sejak Agustus 2007.
- a. Seandainya tidak ada kebijakan yang diambil oleh pemerintah maupun otoritas moneter di Indonesia, apakah yang akan terjadi pada nilai tukar riil rupiah terhadap dolar Amerika? Jelaskan menggunakan diagram yang tepat.
 - b. Apa yang akan terjadi pada nilai ekspor-impor di Indonesia jika terjadi apresiasi/depresiasi nilai tukar riil rupiah pada soal (a)? Jelaskan.